

ENTERPRISE AI SUMMIT · CHARLOTTE, NORTH CAROLINA

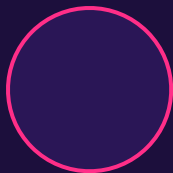
Scaling Confidence Through AI-Powered Resiliency Testing

Christina Yakomin

Senior Manager @ Vanguard

THE TENSION

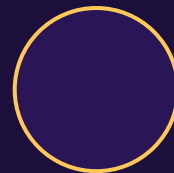
Confidence Cannot Become a Casualty of Speed



Vibe coding is here



Code output is soaring



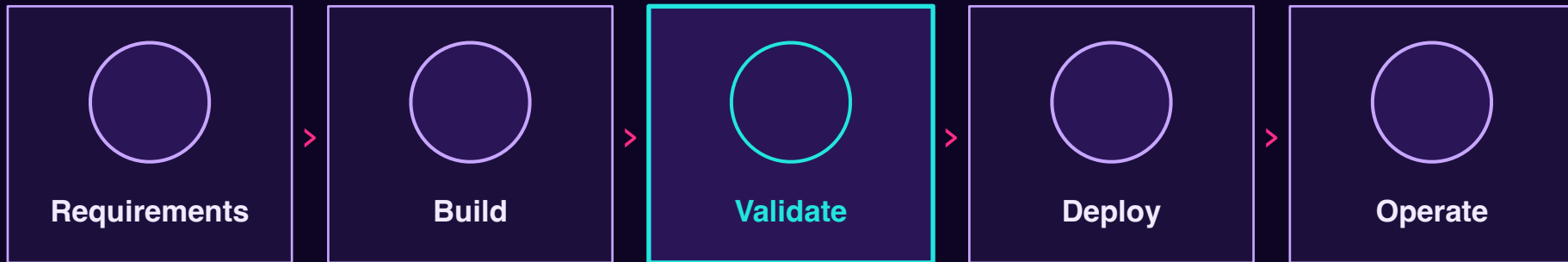
Validation must keep
pace

Accelerate delivery and increase rigor — at the same time.

FUTURE-STATE VISION

A Platform for Builders

AUTOMATION ACROSS EVERY STAGE

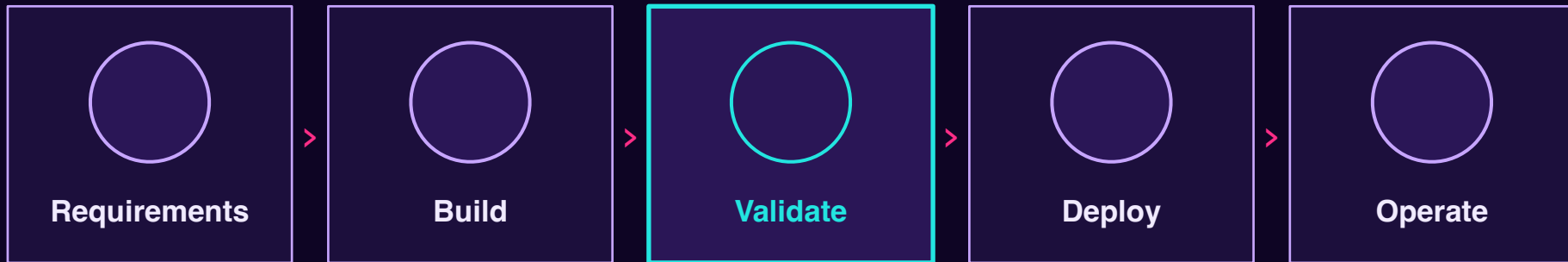


Close the gap between CI build and CD release.

FUTURE-STATE VISION

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AUTOMATION ACROSS EVERY STAGE



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FUTURE-STATE VISION

Does Your Code Pass the Vibe Check?



Performance



Reliability



Efficiency



Operability



Security



Accessibility

Performance + Reliability + Efficiency · today's deep dive

One quality capability. Every non-functional requirement. Continuously.

What We Actually Mean by Each Pillar

Performance

Validate that an application can meet expected workload demands.

- Can it handle expected traffic volumes?
- How does response time change under load?

Reliability

Validate that an application can withstand and recover from failures.

- What happens when dependencies fail?
- Can it recover automatically?

Efficiency

Validate that an application achieves its objectives with acceptable resource consumption.

- Are cloud resources appropriately sized?
- Has a change introduced unnecessary cost?

Every pillar answers a different question about production readiness.

The Pillars We're Building Toward

Operability

Validate that an application can be effectively monitored, diagnosed, and supported in production.

- Will signals identify issues quickly?
- Can issues be caught before clients feel them?

Security

Validate that an application's security controls effectively resist malicious behavior.

- Can an attacker bypass security controls?
- Can sensitive data be exposed?

Accessibility

Validate that an application can be used by people with a wide range of abilities, assistive technologies, and interaction methods.

- Can key tasks be done with a screen reader?
- Can it be operated keyboard-only?

Same model. Different evidence. Same goal - production readiness, proven.

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BY THE NUMBERS

We Scaled Testing, Then Changed How We Measured Success

2024

~20,000

resiliency tests

>

2025

~50,000

resiliency tests

>

2026

Test Quality

became the measure

Volume matters, but quantity without quality doesn't tell the whole story.

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BY THE NUMBERS · FIRST HALF 2026

The Metrics Revealed a Confidence Gap

~50%

of runs were valid

<5%

achieved their goals

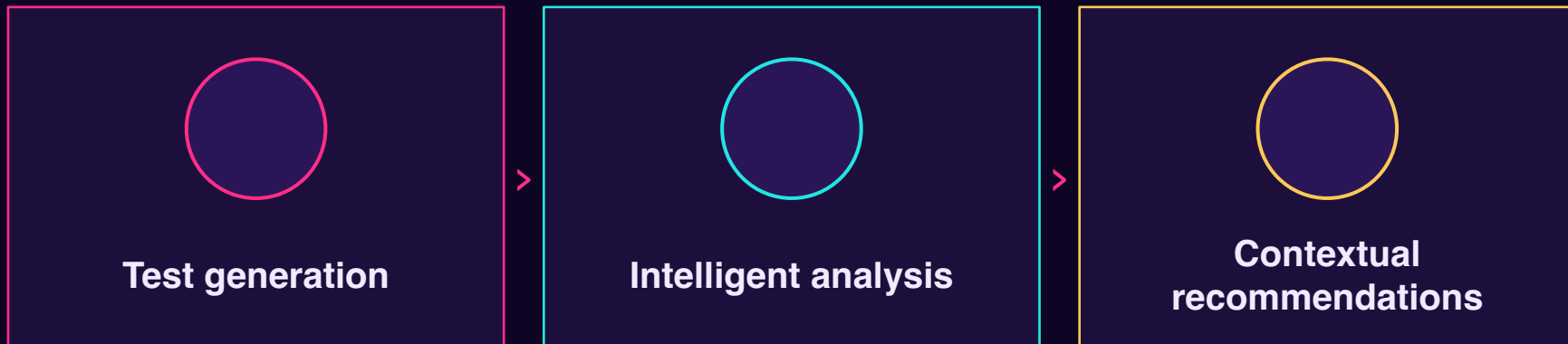
<2%

of production
deployments tested

These signals motivated a new approach.

THE APPROACH

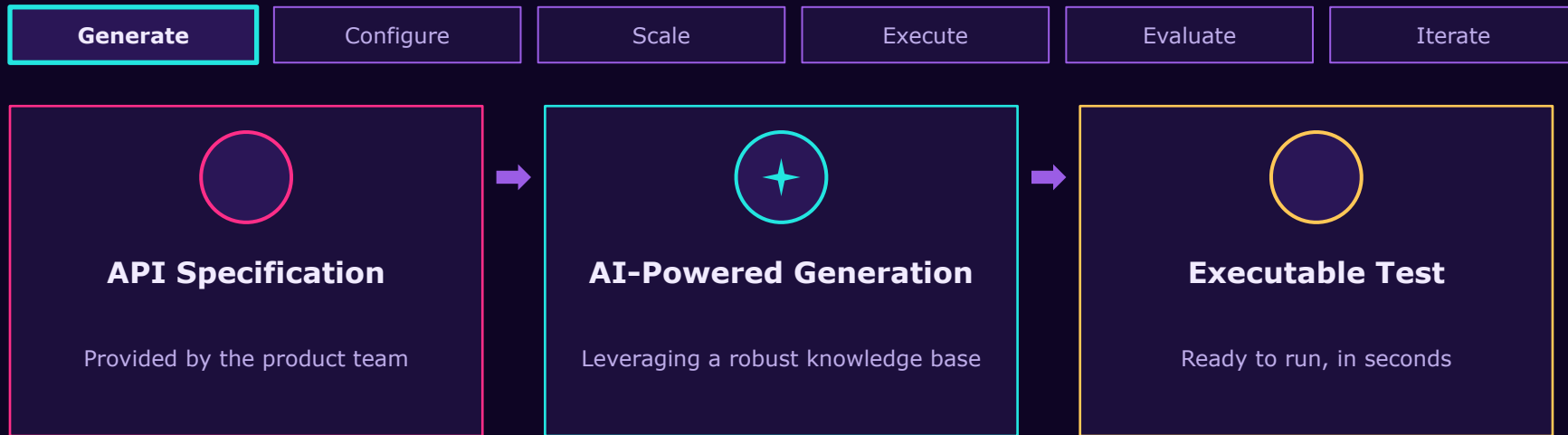
AI Can Reduce the Effort Required to Test Well



By lowering the effort, effectiveness and coverage both rise.

TEST GENERATION

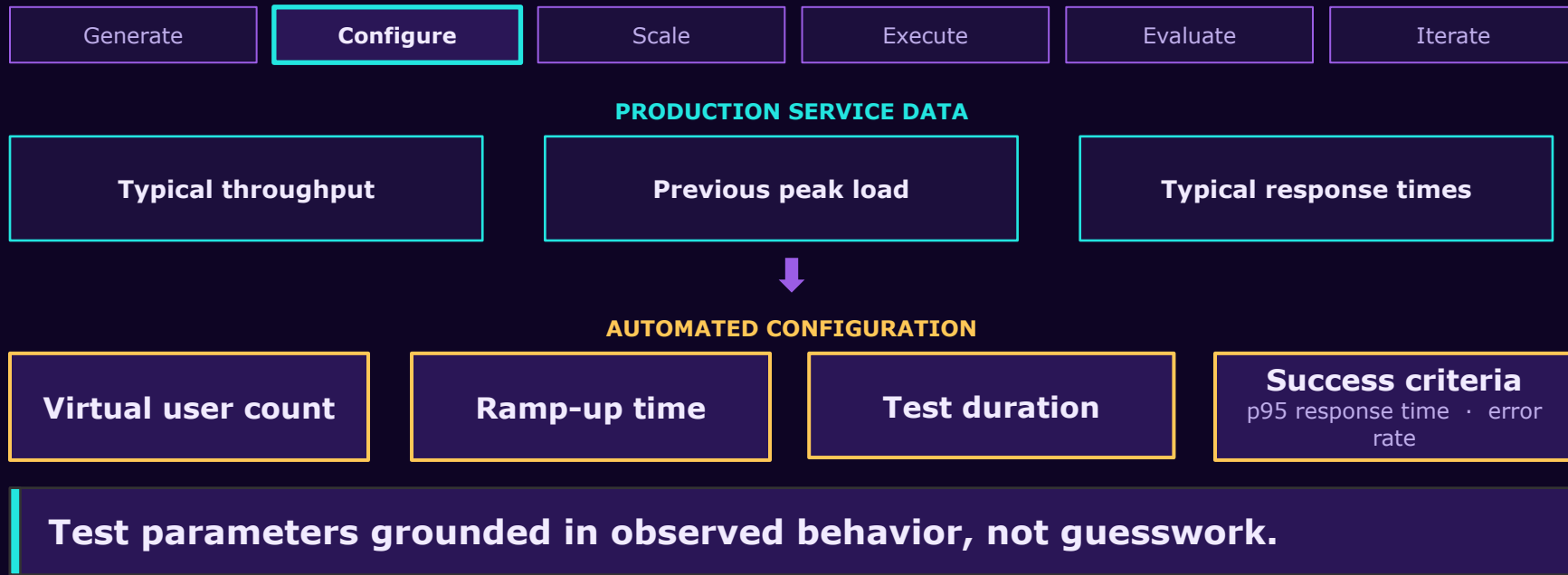
From an API Specification to an Executable Test



No scripting expertise required to start testing.

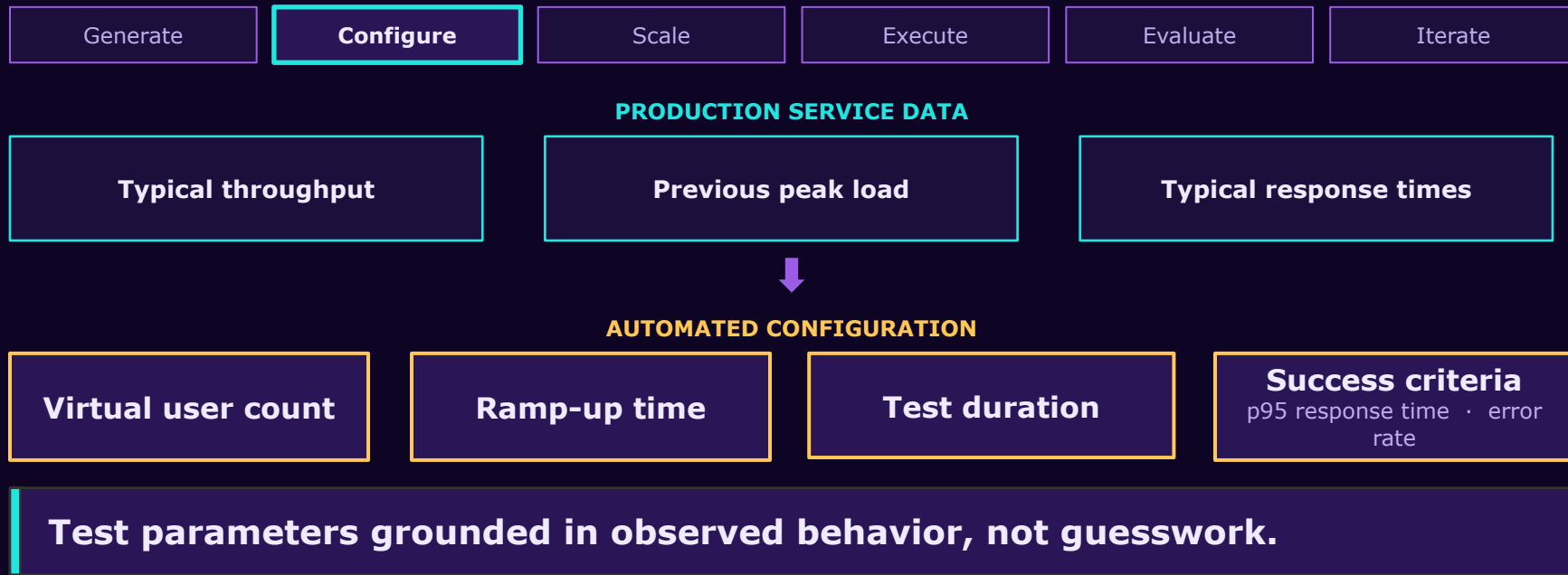
TEST CONFIGURATION

Production Data Determines How the Test Should Run



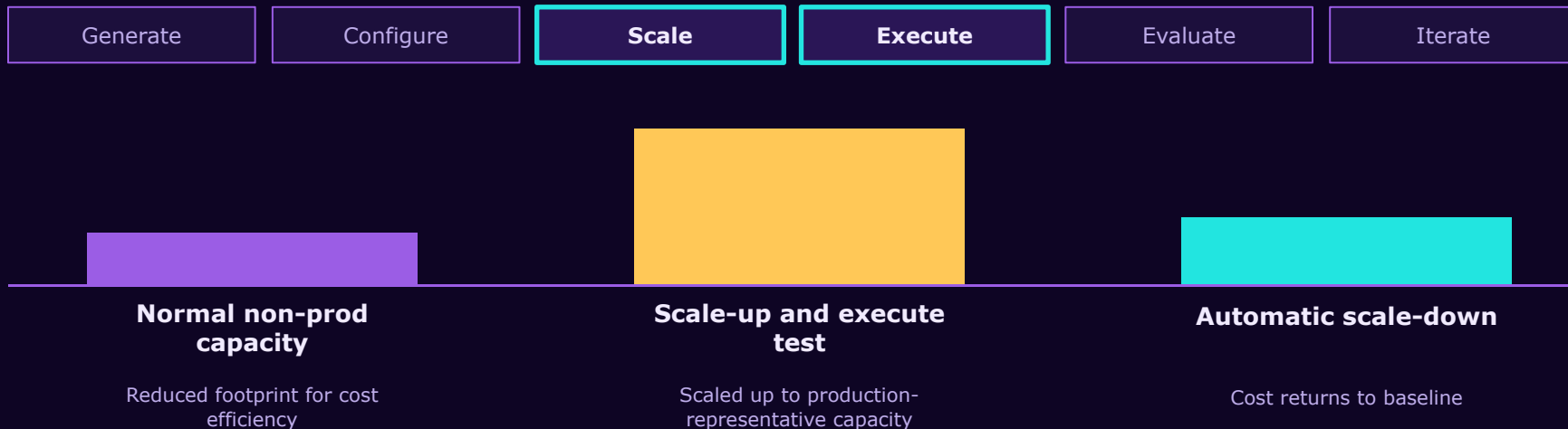
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SCALING AND EXECUTING TESTS

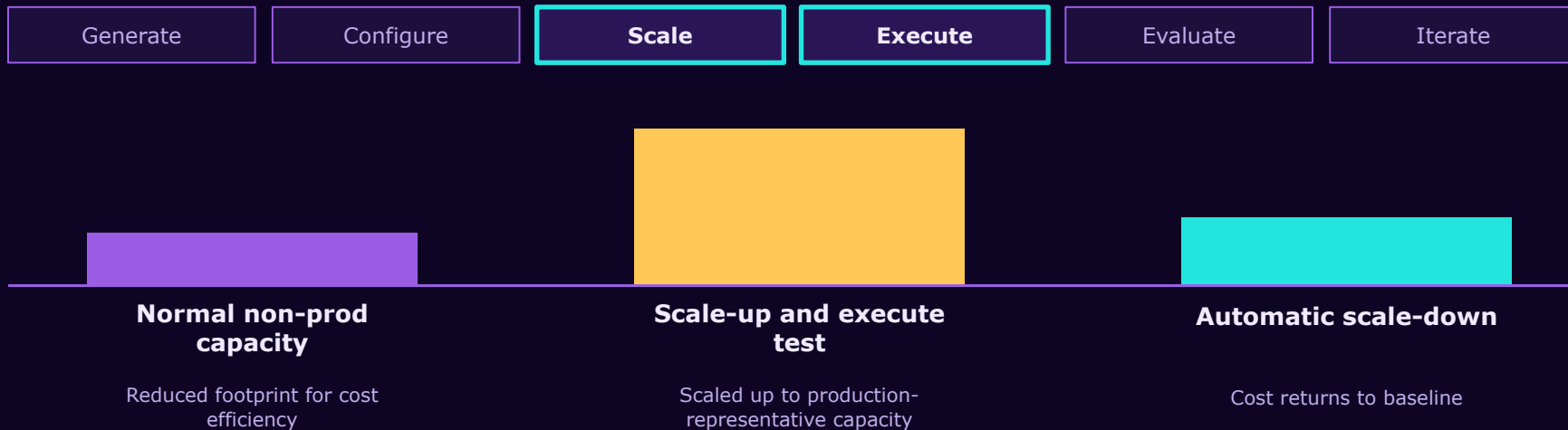
Production-Like Scale Without Permanent Production-Like Cost



Representative results without a permanently oversized environment.

SCALING AND EXECUTING TESTS

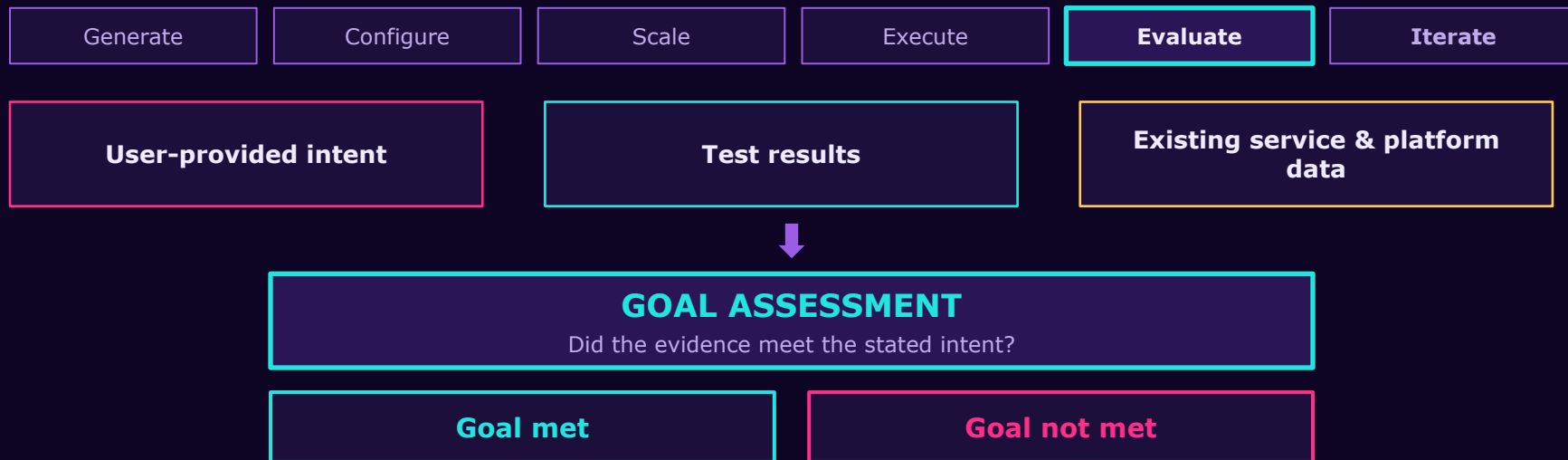
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EVALUATING RESULTS

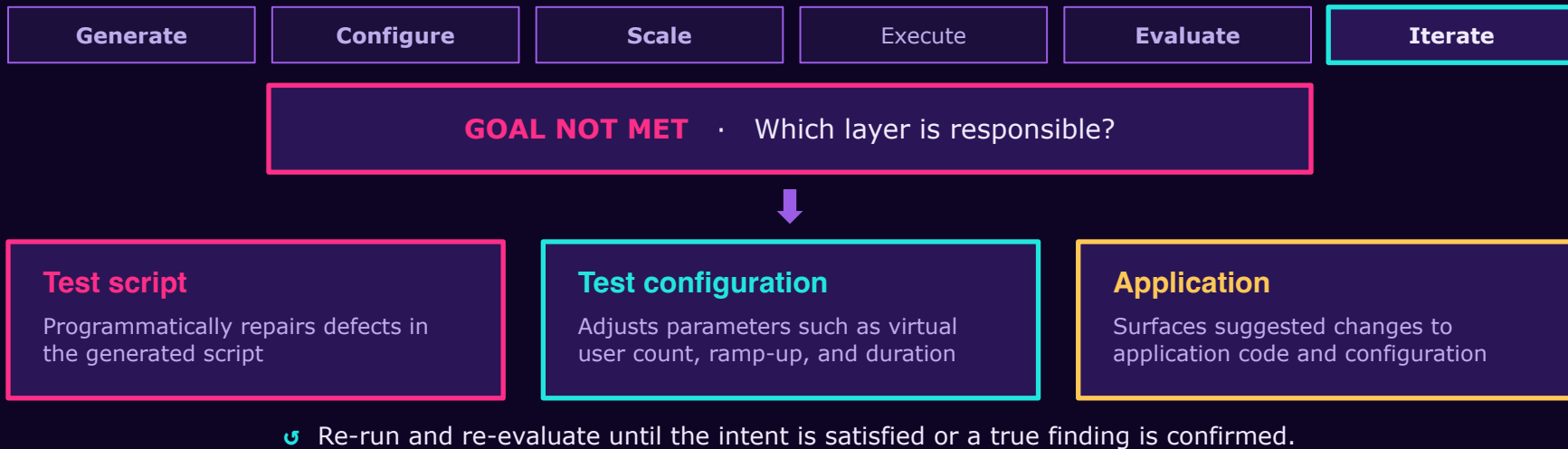
Test Intent Turns Results Into an Assessment



Executing a test is not the same as achieving its purpose.

AUTOMATING ITERATION

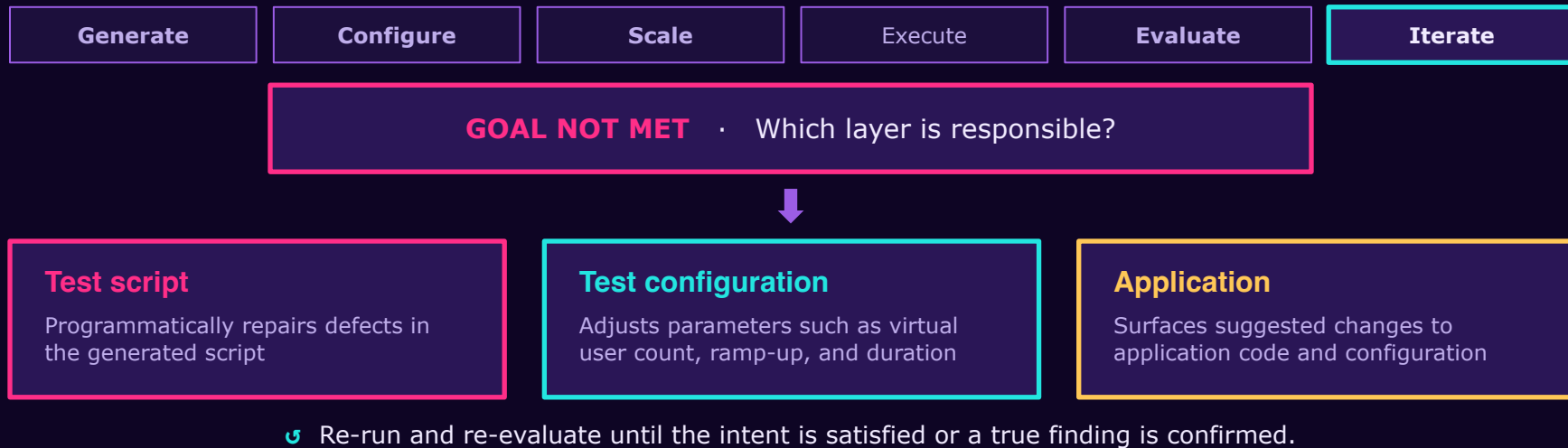
When the Goal Isn't Met, the Agent Doesn't Just Report It



Remediation, not just reporting.

AUTOMATING ITERATION

When the Goal Isn't Met, the Agent Doesn't Just Report It



Remediation, not just reporting.

BY THE NUMBERS · SINCE JULY 2026

Early Indicators of Success

~75%

of runs were valid

+50%

tests achieving their
goals

+10%

overall test volume year-
over-year

Intelligent automation has driven immediate improvements to test effectiveness.

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WHERE THIS GOES NEXT

Today's Resiliency Automation Points to Tomorrow's Vibe Check

Software Release
Built



VIBE CHECK

Determines applicable requirements and the evidence required



SPECIALIZED AGENTS · generate, execute, analyze, and iterate within their domain

Performance

Reliability

Efficiency

Operability

Security

Accessibility



Requirements not met → iterate

Requirements met → Prod Ready

Resiliency today. Orchestrated quality tomorrow.

FUTURE-STATE VISION

HELP I NEED

How are YOU Checking the Vibe?



Performance



Reliability



Efficiency



Operability



Security



Accessibility

I need your help to bring this vision to life!

SCALING EXPERTISE

We Never Had Enough SREs to Go Around

The goal was never to replace them. It was to amplify them.

THE EXPERT'S TIME, BEFORE

Supporting one team at a time

Building a test, tuning a configuration,
interpreting a result

Depth limited by how many teams one
person could reach

THE EXPERT'S TIME, NOW

Defining what good resiliency testing looks
like as architectures evolve

Teaching the system to recognize new
failure modes

Finding the gaps automation can't see yet

Their roles didn't get smaller. Their expertise got distributed.

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THE TENSION, RESOLVED

Confidence cannot be a casualty of speed.

**Scalable confidence,
powered by AI & automation.**

More rigor

More speed

No tradeoffs

Christina Yakomin · Senior Manager @ Vanguard